

John Abrahams

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science

Expected December 2026

M.S.E. in Data Science · GPA 3.62/4.0

Coursework: Theory of Machine Learning, Emerging Theory of Large Language Models, Advanced Operating Systems

Lehigh University, P.C. Rossin College of Engineering and Applied Science

May 2025

B.S. in Computer Science and Engineering · GPA 3.65/4.0 (major 3.75) · Minor: Probability and Statistics

EXPERIENCE

Ninth Wave, Software Engineer Intern, AI Team

May 2026 – Aug 2026

- Built a generator that turns one YAML file per bank into that bank's AWS stacks. Onboarding had meant about ten hand edits per bank, and all three production banks had shipped with a routing defect.
- Built the pipelines behind self-updating documentation: a weekly Lambda that reads the AWS accounts and rewrites the docs with Bedrock, and a scanner that drafts updates when code changes. Tested but not yet deployed.
- Built Studio, a gateway that lets an AI agent call any internal app through one MCP endpoint, with one set of rules for who may do what. Rewrote it after security review found two ways a caller could act as someone else.
- Shipped eleven changes to the Spring Boot portal that banks log into: an activity timeline with channels and search, a browser for the files banks keep in S3, and documentation white-labeled per bank.

New York Brain & Spine Surgery, Software Engineer

May 2025 – Mar 2026

- Built a production therapy chatbot on LLaMA and FastAPI. Clinical rules control the flow of each conversation and the model writes the replies, so one system covers several conditions.
- Co-authored a study of outcomes for 236 spinal-fusion patients as a research associate there in 2023.

RESEARCH

Kronos: WebAssembly Runtime, UPenn (advisor: Prof. Sebastian Angel)

Oct 2024 – Present

- Built a WebAssembly runtime in Rust that runs many tenants in one process. Each tenant costs about 0.08 MiB of memory, about a twentieth of a Docker container, and gets its own CPU and wall-clock budget.
- Wrote a work-stealing thread pool for guest threads. Matrix multiply runs up to 6.7x faster on four cores.
- Co-wrote a nine-page paper on the design with Ajax Li (unpublished).
- Turned the runtime into kronoslabs.dev, a live site where a plain-language request becomes a sandboxed app.

Learning After Deployment, independent research

Sep 2026 – Present

- Studying how a deployed language model should learn new facts without forgetting what it already knows. The method: train the model to reproduce what it would have said with the fact in its prompt.
- Found on Pythia-410M that this stores the fact itself rather than the sentence it came from, and forgets about a third as much as ordinary fine-tuning at the same recall. The result holds at 1B.
- Added a penalty for drift on unrelated text, which brought forgetting to near zero while the model still answered 97% of questions about the new facts. Synthetic facts only so far; unpublished.

Recurrent Reasoning Grafts in Frozen Language Models, independent research

Jul 2026 – Sep 2026

- Tested whether adding a small recurrent reasoning module inside a frozen language model beats plain LoRA fine-tuning on multi-hop questions. It does not: LoRA won every test by 0.4 to 0.85 in accuracy across three seeds.
- Ran the control that could have rescued the idea: trained from scratch, the same blocks score higher when run once than when run in a loop. Wrote it up as a draft paper, unpublished.

PROJECTS

DJ Copilot, independent audio-ML project

Jul 2026 – Present

- Embedded 42,397 tracks with a music transformer (MERT) and trained a model to predict the next track in a DJ set from 1.07M examples. It lost to a simple count of what usually follows each track (4.6% vs 6.9% recall@10).
- Added features for what the DJ community was playing each month, which more than doubled recall@10 for tracks with no play history, from 3.1% to 7.8%. Shuffling the months erased the gain, so the effect is timing, not fame.

SKILLS

Languages: Python, Rust, TypeScript, Java, C++, Go

Machine Learning: PyTorch, LoRA fine-tuning, contrastive learning, reinforcement learning, MERT

Systems: WebAssembly, AWS CDK, Spring Boot, FastAPI, MCP, AWS (Bedrock, S3, Lambda), Kubernetes (K3s)